Week 12 — Data-Driven Two-Stage LAP 1

OL7014 Integer Programming • Instructor: Prof. H. Park • Week of Nov 9, 2026

1. Uncertain data — decide now, see later

week 1: every number known. today: some data is **random** — call it $\tilde{\xi}$, $\tilde{\xi} \sim \mathbb{P}$; a realization is written ξ

Deterministic program (week 1)

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned}$$

Stochastic program

$$\boxed{\begin{aligned} \min_{x} \quad & \mathbb{E}_{\mathbb{P}}\big[\, f(x,\tilde{\xi}) \,\big] \\ \text{s.t.} \quad & x \in X \\ & \mathbb{P}\big(\, g(x,\tilde{\xi}) \le 0 \,\big) \;\ge\; 1-\varepsilon \end{aligned}} \qquad \text{(SP)}$$

2. The chance constraint — a MIP

$$\mathbb{P}\big(\, g(x,\tilde{\xi}) \le 0 \,\big) \;\ge\; 1-\varepsilon \qquad\Longleftrightarrow\qquad \mathbb{P}\big(\, g(x,\tilde{\xi}) > 0 \,\big) \;\le\; \varepsilon$$

take $\mathbb{P}$ = the data: N equally likely scenarios $\xi^1, \dots, \xi^N$ $\Rightarrow$ the probability is a **count**:

$$\frac{1}{N}\sum_{n=1}^{N} \mathbb{I}\big[\, g(x,\xi^n) > 0 \,\big] \;\le\; \varepsilon \qquad \text{— at most } \varepsilon N \text{ scenarios may fail}$$

one binary per scenario (week 2's big-M, week 4's trick): $z^n = 1 \iff$ scenario n may violate

$$g(x,\xi^n) \;\le\; M z^n \qquad z^n = 0 \;\Rightarrow\; g(x,\xi^n) \le 0; \qquad z^n = 1 \;\Rightarrow\; \text{row says nothing} \quad (M \ge \max g)$$

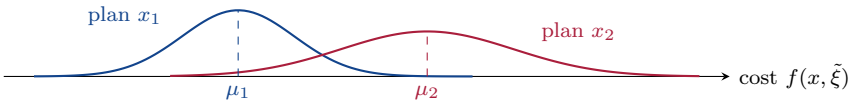
the MIP:

$$\begin{aligned} \min_{x,\,z} \quad & f(x) \\ \text{s.t.} \quad & x \in X \\ & g(x,\xi^n) \le M z^n && \forall n \in [N] \\ & \textstyle\sum_{n=1}^{N} z^n \le \varepsilon N, \quad z^n \in \{0,1\} && \forall n \in [N] \end{aligned}$$

$N = 100$, $\varepsilon = 0.05$: at most 5 scenarios fail — *and the solver chooses which* 5

3. Why the mean — every plan has a cost distribution

fix x $\Rightarrow$ $f(x,\tilde{\xi})$ is a random number — it has a *distribution*



which plan is better? compare the means: $\mu_1 < \mu_2 \;\Rightarrow\; x_1$. that is (SP): $\min_{x\in X} \mathbb{E}_{\mathbb{P}}[\, f(x,\tilde{\xi}) \,]$

watch the mean is one way to rank distributions. others: worst case, tail (CVaR). not today.

4. SAA — replace $\mathbb{P}$ by the data

$\mathbb{P}$ is unknown. what we have: N past realizations of $\tilde{\xi}$: $\xi^1, \xi^2, \dots, \xi^N$ — the **scenarios**

Def. Sample Average Approximation (SAA):

$$\underbrace{\min_{x\in X} \; \mathbb{E}_{\mathbb{P}}\big[\, f(x,\tilde{\xi}) \,\big]}_{\text{true problem — cannot compute}} \qquad \approx \qquad \underbrace{\min_{x\in X} \; \frac{1}{N}\sum_{n=1}^{N} f(x,\xi^n)}_{\text{SAA problem — every number known again}}$$

why: for a fixed x, $\frac{1}{N}\sum_n f(x,\xi^n) \to \mathbb{E}_{\mathbb{P}}[\, f(x,\tilde{\xi}) \,]$ as $N \to \infty$ (law of large numbers)

classification: $\xi^n = (\boldsymbol{x}_n, y_n)$, the N labelled points

$$\min_{\theta} \; \frac{1}{N}\sum_{n=1}^{N} \mathbb{I}\big[\, y_n\,(\theta^\top \boldsymbol{x}_n) \le 0 \,\big] \qquad \text{— week 4's model, exactly: week 4 } \textit{was} \text{ SAA}$$

"plug in the average" here would mean one point $(\bar{\boldsymbol{x}}, \bar{y})$ — meaningless

takeaway **SAA is *not* "plug in the average" — it keeps every data point.**

5. Two stages — the cost is itself an optimization



single stage	$f(x,\xi)$ = a formula	nothing left to decide after ξ
two stages	$f(x,\xi) = c^\top x + Q(x,\xi)$	after ξ we still choose y

stage 1: $x \in X$ — week 2's MILP constraints, $X = \{\, x \in \mathbb{Z}^n_+ : Ax \le b \,\}$. stage 2: new letters q, T, W

Def. the **recourse function** — the best reaction, once x is fixed and ξ is seen:

$$Q(x,\xi) \;=\; \min_{y} \left\{\, q^\top y \;:\; Tx + Wy \ge \xi, \quad y \ge 0 \,\right\} \qquad \xi = \text{the right-hand side}$$

(in general $\tilde{\xi}$ can be *all* the stage-2 data (q,T,W,h) — today only the right-hand side is random)

same model, same SAA:

$$\min_{x\in X} \; c^\top x + \mathbb{E}_{\mathbb{P}}\big[\, Q(x,\tilde{\xi}) \,\big] \qquad \overset{\text{SAA}}{\Longrightarrow} \qquad \min_{x\in X} \; c^\top x + \frac{1}{N}\sum_{n=1}^{N} Q(x,\xi^n)$$

write every Q out — one big MILP, the **extensive form**:

$$\min_{x,\; y^1,\dots,y^N} \; c^\top x + \frac{1}{N}\sum_{n=1}^{N} q^\top y^n \qquad \text{s.t.} \quad Tx + Wy^n \ge \xi^n \;\; (n=1,\dots,N), \quad x \in X, \quad y^n \ge 0$$

$N = 3$:

$$\begin{bmatrix} T & W & & \\ T & & W & \\ T & & & W \end{bmatrix} \begin{bmatrix} x \\ y^1 \\ y^2 \\ y^3 \end{bmatrix} \;\ge\; \begin{bmatrix} \xi^1 \\ \xi^2 \\ \xi^3 \end{bmatrix} \qquad \text{one shared } x;\;\; \text{one block per scenario;}\;\; \text{plus } x \in X$$

takeaway **more data ⇒ a bigger MILP: N copies of y, N copies of the rows.**

6. The two-stage LAP

week 6: $d = (12, 20, 8, 16, 24)$ was *known* when we chose where to build. really?

the demand is random: $d \to \tilde{\xi}$, ξ^n_i = demand at node i in scenario n

stage 1 — before the demand	x_i open, r_i stock	no n
stage 2 — after ξ^n is seen	y^n_{ij} ship, q^n_i unmet, s^n_i surplus	one copy per scenario

week 6's model, with $d \to \xi^n$ and a superscript n on stage 2:

$$\min \quad \underbrace{\textstyle\sum_{i\in V} f_i x_i + \sum_{i\in V} h_i r_i}_{\text{stage 1}} \;+\; \frac{1}{N}\sum_{n=1}^{N} \underbrace{\Big[\, \textstyle\sum_{(i,j)\in E} c_{ij}\, y^n_{ij} + \sum_{i\in V} u_i\, q^n_i + \sum_{i\in V} o_i\, s^n_i \,\Big]}_{\text{stage 2, scenario } n}$$

$$\text{s.t.} \quad \textstyle\sum_j y^n_{ij} - \sum_j y^n_{ji} = r_i - \xi^n_i + q^n_i - s^n_i \qquad (\forall i \in V,\; \forall n) \quad \text{[conservation, per scenario]}$$

$$\textstyle\sum_{i\in V} r_i \le B, \qquad r_i \le B\, x_i \;\; (\forall i \in V), \qquad \sum_{i\in V} x_i \le p \qquad \text{[stage 1: as in week 6]}$$

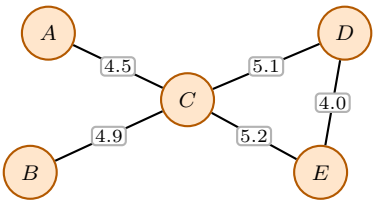
$$y^n, s^n \ge 0, \quad 0 \le q^n_i \le \xi^n_i, \quad r \ge 0, \quad x_i \in \{0,1\}$$

$N = 1$: week 6's model, exactly

fix (x,r) ⇒ N separate MCNFs (week 5) — the hardness is still in x

watch r is continuous but first-stage: the split is now by *time*, not by integer vs. continuous.

7. Numerical example — week 6's map, three futures



n	ξ^n_A	ξ^n_B	ξ^n_C	ξ^n_D	ξ^n_E	total	
1	12	20	8	16	24	80	week 6's d
2	30	25	5	0	0	60	west
3	0	0	5	25	30	60	east

week 6's map: $f_i = 100$, $h_i = 10$, $u_i = 60$, $o_i = 1$, $p = 2$, $B = 55$

$N = 3$:

$$\min \quad 100\sum_i x_i + 10\sum_i r_i \;+\; \frac{1}{3}\sum_{n=1}^{3}\Big[\,4.5\,(y^n_{AC}+y^n_{CA}) + \cdots + 4.0\,(y^n_{DE}+y^n_{ED}) + 60\sum_i q^n_i + \sum_i s^n_i\,\Big]$$

$$\begin{aligned}
\text{s.t.}\quad & y^1_{AC} - y^1_{CA} = r_A - 12 + q^1_A - s^1_A && \text{[node } A\text{, scenario 1]}\\
& y^2_{AC} - y^2_{CA} = r_A - 30 + q^2_A - s^2_A && \text{[node } A\text{, scenario 2]}\\
& y^3_{AC} - y^3_{CA} = r_A - \;\;0 + q^3_A - s^3_A && \text{[node } A\text{, scenario 3]}\\
& \vdots && \text{[nodes } B, C, D, E\text{: 15 rows in all]}\\
& \textstyle\sum_i r_i \le 55, \qquad r_i \le 55\,x_i, \qquad \sum_i x_i \le 2 && \text{[stage 1]}
\end{aligned}$$

the same r_A in all three rows

size: stage 1: $5 + 5 = 10$ variables; each scenario: $10 + 5 + 5 = 20$; total $10 + 20N = 70$

solve:

plan built for	opens, stock r	stage 1	stage-2 cost in scenario 1	2	3	**average total**
scenario 1 only (week 6)	B: 20, E: 35	750	1544	617	494.5	1635.2
scenario 2 only	A: 30, B: 25	750	1682	300	815	1682.3
scenario 3 only	D: 25, E: 30	750	1608.6	816.5	300	1658.4
all three (SAA)	C: 55	650	1728.5	533	557.5	**1589.7**

average total = stage 1 + $\frac{1}{3}$(three stage-2 costs)

takeaway **SAA opens C — no single future would. best in no scenario, best on average.**

demo `LAP-Dashboard.exe` — Solve, then Simulate disaster